Internal Assessment 1 - Chemistry

Semester II | Time: 2 hours | Maximum marks: 50

Answer ALL questions. Each question carries 5 marks. Draw neat, labelled diagrams in every answer.

Q1. Atomic structure **[5 marks]**

Draw the Bohr model of a sodium atom (Z = 11, A = 23) showing the number of protons and neutrons and the arrangement of electrons in shells. Write its electronic configuration and valency.

Q2. Shape of the water molecule **[5 marks]**

Draw the electron-dot (Lewis) structure of H2O and a diagram of its shape. State the H-O-H bond angle and explain why it is less than the tetrahedral angle of 109.5°.

Q3. Ionisation energy **[5 marks]**

Sketch a graph of first ionisation energy against atomic number for the period 2 elements (Li to Ne). Explain the general trend and the drops at boron and oxygen.

Q4. Daniell cell **[5 marks]**

Draw a labelled diagram of the Daniell cell. Write the anode and cathode half-reactions and calculate the standard cell EMF. (E° Zn2+/Zn = -0.76 V, E° Cu2+/Cu = +0.34 V)

Q5. Catalysis **[5 marks]**

Draw an energy profile diagram for an exothermic reaction with and without a catalyst. Label the activation energies and ΔH, and explain how a catalyst increases the rate of reaction.

Q6. Simple distillation **[5 marks]**

Draw a labelled diagram of the apparatus for simple distillation. Explain the purpose of the condenser and why the cooling water enters at the bottom.

Q7. Acid-base titration **[5 marks]**

25.0 mL of 0.10 M HCl is titrated with 0.10 M NaOH. Sketch the titration curve (pH against volume of NaOH added) and mark the equivalence point. Calculate the volume of NaOH at the equivalence point and name a suitable indicator.

Q8. Benzene **[5 marks]**

Draw the two Kekulé structures and the resonance hybrid of benzene. Explain why benzene prefers substitution reactions to addition reactions.

Q9. Boyle's law **[5 marks]**

State Boyle's law and sketch a graph of pressure (P) against volume (V) for a fixed mass of gas at constant temperature. A gas occupies 3.0 L at 2.0 atm. Find its volume at 6.0 atm at the same temperature.

Q10. Electrolysis **[5 marks]**

Draw a labelled diagram for the electrolysis of molten sodium chloride. Write the reactions at the cathode and the anode and name the products formed.